

1. What is the role of a FC in a drone system, and how does it differ from an onboard system like Jetson?

A. A. FC (Flight Controller)

- This acts like the autonomous nervous system
- It maintains drone stability using sensor data(IMU, GPS, barometer)
- It processes radio inputs/executes control loops
- It's not usually free-acting it relies on external commands via MAVLink/UART/CAN

B. Onboard Computer

- It's the somatic nervous system
- It handles more advanced tasks like computer vision, path planning, autonomy
- It runs ROS nodes, SLAM, object detection, and missile logic
- It sends commands to the flight controller and receives vital telemetry for making decisions
- It connects with Ethernet or USB depending on the setup

2. What sensors are typically inside/connected to a flight controller?

A. IMU

- It combines accelerometers/gyroscopes to measure orientation, angular velocity, and linear acceleration
- This is needed for flight stabilization/attitude control
- This is fused with magnetometers for heading estimation

B. Magnetometer

- This detects Earth's magnetic field to determine heading
- This corrects yaw drift/supports navigating in GPS-denied environments
- This is built in to the IMU package but it sucks to calibrate

C. Barometer

- Measures Air Pressures in Earth (atmosphere pressures to estimate altitude)
- It is used for altitude hold/smooth vertical control

D. GPS Module(s)

- This provides the global position, velocity, and time data
- This allows us to navigate waypoints, return to home, and do geofencing!

On our drone we have Standard RGB cameras

- These can send optical data to the FC/optical Computer
- This captures video and full-color images

- This is typically used for mapping/inspection/aerial photography
- This is often mounted on gimbals to stabilize footage

Key Specs

- The bigger sensor means better low-light performance (size)
- Resolution (video? pixels)
- FOV, frame rate, and dynamic range
- Interface type(HDMI, USB, Ethernet, UART)
- Digital vs optical zoom

Rc Transmitter (TX)

- If we manually fly it we use a handheld controller to send commands
It will read inputs and switch positions
It communicates free-wire with receiver with radio protocols

RC Receiver (RX)

- Stationed on drone, receives transmitter signals
- It gives commands to flight controller via protocols like SBUS

3. What are the key communication protocols between flight controllers and peripherals (I2C, CAN, MAVLink)

A. Serial (UART)

- Direct, basic communication between EXACTLY 2 devices
- It facilitates communication between flight controller/onboard

- It has an easy 2-wire setup: TX(transmit) and RX (receive)
Each UART connection has its own wires, it is common for MAVLink messages and debugging

B. CAN (Controller Area Network)

- Think of a group chat for devices
- It is very sophisticated used in cars/high-end drones
- If an environment is volatile it is super useful (also long-cable runs)
- It is used for smart ESCs/gimbals/advanced sensor networks
- It has error checking built-in to prevent miscommunications

MAVLINK: software not hardware

MavLink

- Drone language for commands/telemetry
- It isn't a physically done protocol it's a message format that runs over UART/UDP or CAN
- It sends commands and receives data
- It is a powers communication between flight controller/onboard computer

DDS: Mavlink but inside the drone

- This is a communication system used by ROS 2 / modern drones
 - It has messaging in real-time (public-subscribe)
 - It powers complete connection between flight controllers/onboard computers

- It is amazing for real time performance unlike MAVLINK/UART

A common setup we might have is a Hybrid Mavlink/DDS

- **MavLink:** This talks to our ground station over UDP
- **DDS:** This talks between our flight controller/onboard computer

4. What are the key communication protocols between flight controllers and peripherals (I2C, CAN, MAVLink)

A. What is Flight Controller Firmware?

- It's low-level software that we don't make but runs on the flight computer
- Examples:
 - Ardupilot
 - Betaflight
 - PX4
- Firmware can be set up with hundreds of little settings to optimize flight characteristics
 - Ex: is this a big drone? Or is this a supersonic drone that can do mid air flips
- There is various "flight modes" such as auto-stabilizing, full stick control (acro), and more